

Design and Implementation of a Sense Amplifier-Based Edge-Triggered Register in 180 nm CMOS Technology

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Abstract—This paper presents the design and functional simulation of a sense amplifier-based edge-triggered register in 180 nm CMOS technology using Cadence Virtuoso. The register uses a regenerative sense-amplifier front-end followed by a static latch, enabling fast sampling and robust edge-triggered behavior. Transient simulations verify correct operation and demonstrate very small timing requirements: a clock-to-Q delay of 0.188 ns, a setup time below 5 ps, and a hold time of 18 ps, enabling operation above 5.2 GHz. These results confirm that sense-amplifier-based registers remain highly suitable for multi-gigahertz datapaths and low-swing interconnect receivers.

Index Terms—sense amplifier, flip-flop, setup time, clock-to-Q delay, CMOS

I. INTRODUCTION

High-speed processors and DSPs rely on aggressive pipelining, making register delays a major component of the clock period [1]. Traditional CMOS latches suffer from limited speed and relatively large setup windows. In contrast, sense amplifier-based flip-flops (SAFFs) exploit positive feedback to rapidly resolve input differences and close the sampling window almost immediately after the active clock edge. This leads to very small effective setup times, reduced sensitivity to clock skew, and multi-gigahertz operation.

This work implements and analyzes a classical SAFF topology in a 180 nm CMOS process. The objective is to verify its functionality and extract key timing parameters such as setup time, hold time, and the clock-to-Q delay, and to relate these to the achievable maximum clock frequency.

II. CIRCUIT ARCHITECTURE

A. Sense Amplifier Front-End

The front-end consists of a cross-coupled inverter pair with differential NMOS inputs whose internal nodes (L_1 , L_2) are precharged during the low clock phase. On the rising clock edge, a tail NMOS enables differential evaluation. Any small difference at the inputs produces a current imbalance that discharges one of the internal nodes, and the positive feedback forces rapid regeneration. Because this decision occurs very soon after the clock edge, the sense amplifier exhibits an extremely small effective setup time.

A weak “shorting” NMOS ensures that partially discharged nodes do not remain floating and prevents late input changes from corrupting the stored value.

B. Static Output Latch

The regenerated differential outputs drive a static NAND-based SR latch that converts the analog decision into stable, rail-to-rail Q and $\bar{Q}$ outputs. Additional NMOS devices in the pull-down path improve discharge speed and suppress potential glitches. This back-end stage ensures one output transition per cycle and preserves the decision made by the sense amplifier.

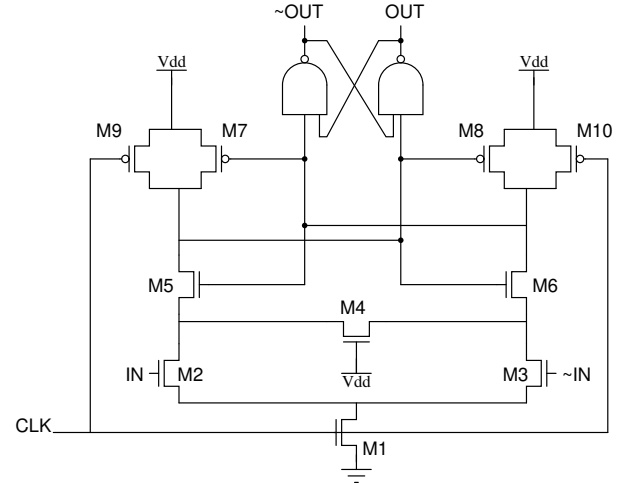


Fig. 1: Schematic of the implemented sense amplifier-based register.

III. SIMULATION SETUP

All simulations were performed in Cadence Virtuoso using a 180 nm CMOS PDK and the Spectre simulator. The supply voltage was fixed at 1.8 V and the temperature at 27 °C.

Two sets of transient simulations were conducted:

- **Sense1:** Functional simulation with full-swing input to verify correct edge-triggered behavior and complementary outputs.
- **Sense2:** Parametric simulation sweeping the input high level from 1.8 V to 0 V in 0.2 V steps to study regeneration time dependence on input amplitude.

IV. RESULTS

A. Functional Verification (Sense1)

Figure 2 shows the input, clock, Q , and $\bar{Q}$ waveforms. The register switches only on the rising clock edge and holds its

state during the low phase, confirming correct positive edge-triggered behavior.

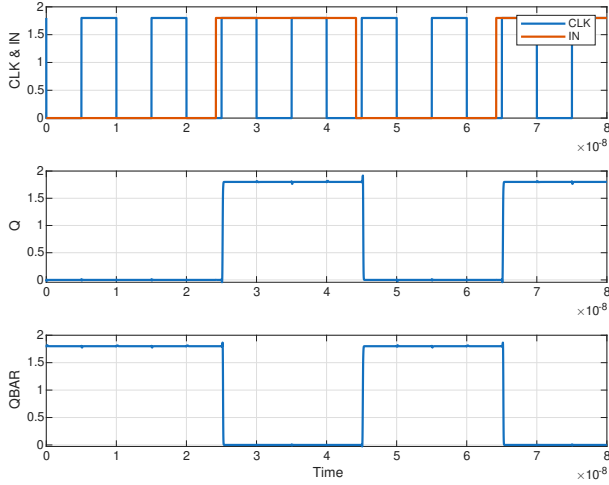


Fig. 2: Sense1 simulation: CLK & IN (top), Q (middle), $\overline{Q}$ (bottom).

B. Regeneration vs. Input Amplitude (Sense2)

Reducing the input high level weakens the initial differential current in the input pair, increasing the evaluation time. Figure 3 shows the Q transitions for decreasing input levels. Smaller amplitudes shift the transition further in time, consistent with sense-amplifier theory.

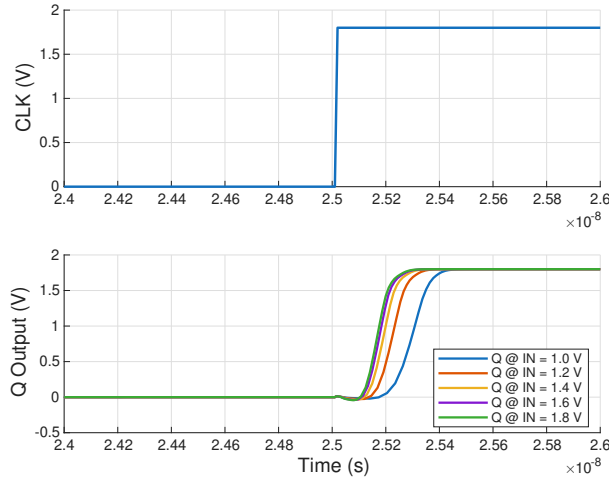


Fig. 3: Sense2 simulation showing Q for input levels from 1.8 V down to 1.0 V.

C. Timing Characterization

Timing was extracted from the Sense1 simulation using the standard 50% rule [1]. The results are:

- Clock-to-Q delay: 0.188 ns
- Setup time: < 5 ps
- Hold time: 18 ps

These correspond to a minimum clock period

$$T_{\min} \approx t_{\text{clk} \rightarrow Q} + t_{\text{su}} \approx 0.193 \text{ ns},$$

and a maximum frequency

$$F_{\max} \approx 5.2 \text{ GHz}.$$

TABLE I: Measured Timing Parameters

Parameter	Symbol	Value
Supply voltage	V_{DD}	1.8 V
Clock-to-Q delay	$t_{\text{clk} \rightarrow Q}$	0.188 ns
Setup time	t_{su}	< 5 ps
Hold time	t_{h}	18 ps
Max. frequency	—	> 5.2 GHz

V. APPLICATIONS

Because the sense amplifier accepts small differential swings and regenerates them quickly, the implemented register is well suited for:

- High-speed pipeline registers in multi-GHz processors.
- Low-swing differential bus and interconnect receivers.
- High-speed counters and timing-critical synchronous logic.

VI. CONCLUSION

A sense amplifier-based edge-triggered register was designed and simulated in 180 nm CMOS. The circuit demonstrated correct operation with a clock-to-Q delay of 0.188 ns and a sub-5 ps setup time. These results correspond to an operating frequency exceeding 5.2 GHz, confirming the suitability of SAFF-based structures for high-speed, low-swing digital systems. Post-layout analysis and power characterization remain future work.

REFERENCES

- [1] J. Rabaey, A. Chandrakasan, and B. Nikolić, *Digital Integrated Circuits: A Design Perspective*, 2nd ed., Prentice Hall, 2003.
- [2] P. M. Bhatasana *et al.*, “Implementation of sense amplifier-based D flip-flop using 0.25 μm and 0.18 μm technology,” *Journal of Electronic Design Technology*, 2013.
- [3] Cadence Design Systems, “Virtuoso Design Environment User Guide,” 2025.